

Index

S.N	Lab Problems	Date	Signature
1.	Write a program to manipulate text on C/C++ using standard graphics functions	2082/02/20	
2.	Write a program to draw graphics primitives using standard function	2082/02/20	
3.	Write a program to animate a car using graphics functions	2082/02/22	
4.	Write a program to animate a simple Aeroplane using graphics function	2082/02/22	
5.	Write a program/ algorithm to import, edit and export a video on some video authoring software. E.g. Premier pro/ Rush	2082/02/23	
6.	Write a step by step program/ algorithm to upload the video to youtube and embed that video on a HTML page	2082/02/24	
7.	Write a program to implement run length coding on given set of strings	2082/02/25	
8.	Write a program to implement Huffman coding on some given characters and their frequencies	2082/02/25	

Lab 1: Write a program to manipulate text on C/C++ using standard graphics functions

Source code:

```
#include <graphics.h>
#include <conio.h>

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    // Set color and style
    setcolor(WHITE);
    settextstyle(3, HORIZ_DIR, 2);

    // Display initial text
    outtextxy(100, 100, "Hello, Multimedia!");

    // Wait for key press
    getch();

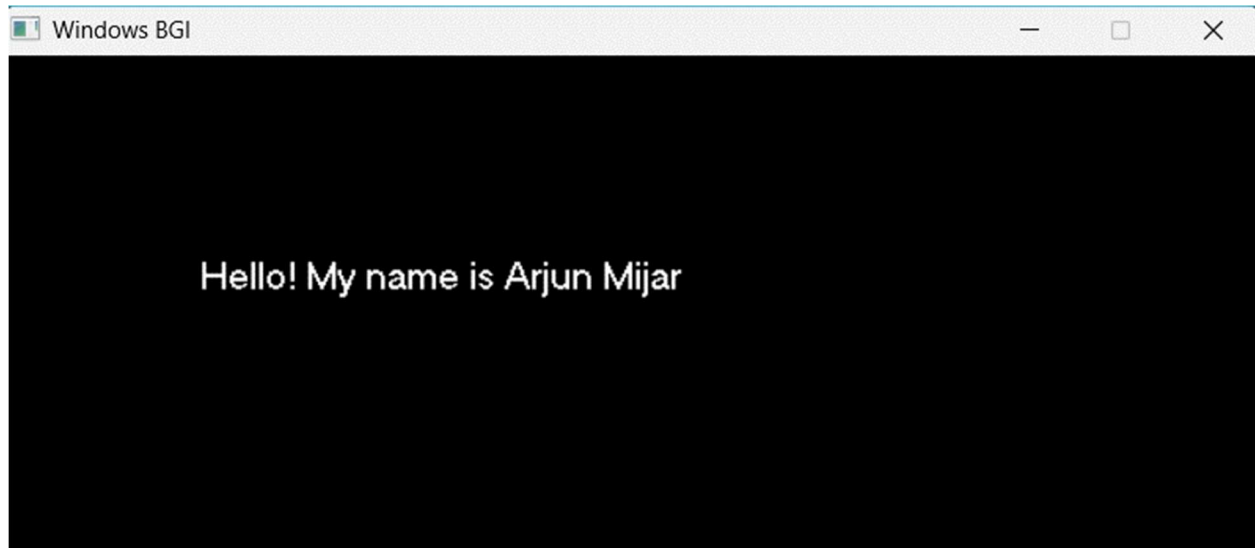
    // Clear screen
    cleardevice();

    // Change color and font
    setcolor(YELLOW);
    settextstyle(2, HORIZ_DIR, 4);
    outtextxy(200, 150, "Text Moved & Styled!");

    // Wait and close
```

```
    getch();  
    closegraph();  
    return 0;  
}
```

Output:



Lab 2: Write a program to draw graphics primitives using standard function.

Source code:

```
#include <graphics.h>  
#include <conio.h>  
int main() {  
    int gd = DETECT, gm;  
    initgraph(&gd, &gm, "");  
    // Draw a line  
    line(100, 50, 300, 50);  
    // Draw a rectangle  
    rectangle(100, 100, 300, 200);
```

```
// Draw a circle
circle(200, 300, 50);

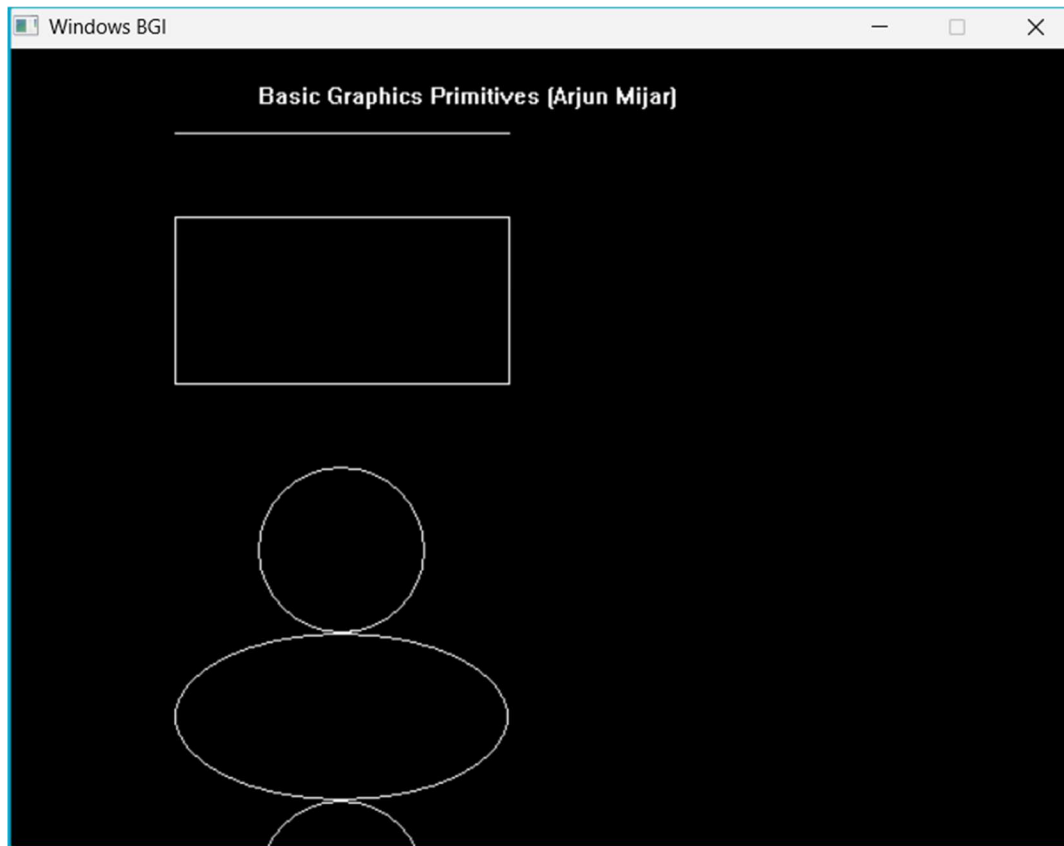
// Draw an ellipse
ellipse(200, 400, 0, 360, 100, 50);

// Draw an arc
arc(200, 500, 0, 180, 50);

// Set text
outtextxy(150, 20, "Basic Graphics Primitives");

// Hold screen
getch();
closegraph();
return 0;
}
```

Output:



Lab 3: Write a program to animate a car using graphics functions.

Source code:

```
#include <graphics.h>
#include <conio.h>
#include <dos.h>

void drawCar(int x, int y) {
    // Car body
    rectangle(x, y, x + 100, y + 20);
    rectangle(x + 20, y - 20, x + 80, y);

    // Wheels
    circle(x + 25, y + 25, 10);
    circle(x + 75, y + 25, 10);
}

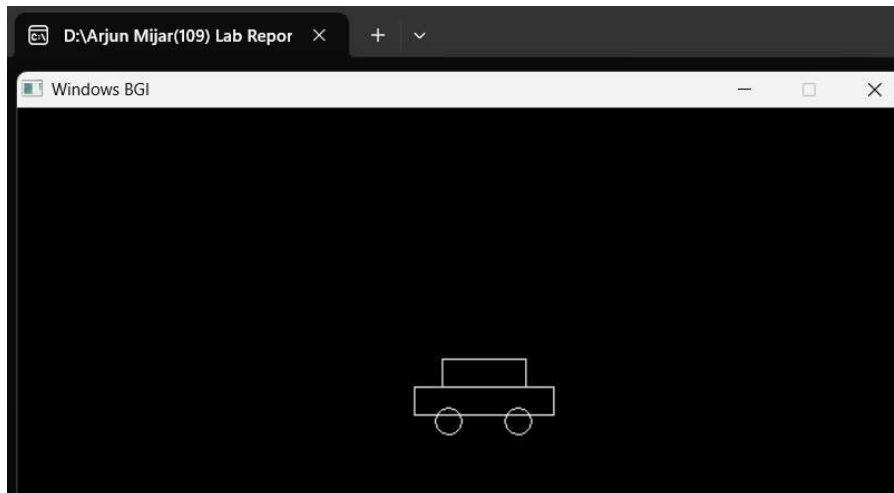
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    for (int i = 0; i < getmaxx(); i += 5) {
        cleardevice();
        drawCar(i, 200);
        delay(50); // Small delay for animation effect
    }

    getch();
    closegraph();
}
```

```
    return 0;
}
```

Output:



Lab 4: Write a program to animate a simple Aeroplane using graphics function

Source code:

```
#include <graphics.h>
#include <conio.h>
#include <dos.h>

// Function to draw a simple aeroplane
void drawAeroplane(int x, int y) {
    // Plane body
    rectangle(x, y, x + 100, y + 20);

    // Tail
    line(x, y, x - 10, y - 10);
```

```

    line(x - 10, y - 10, x, y + 10);

    // Wing
    line(x + 40, y, x + 20, y - 20);
    line(x + 60, y, x + 80, y - 20);

    // Front curve (nose)
    arc(x + 100, y + 10, 270, 90, 10);
}

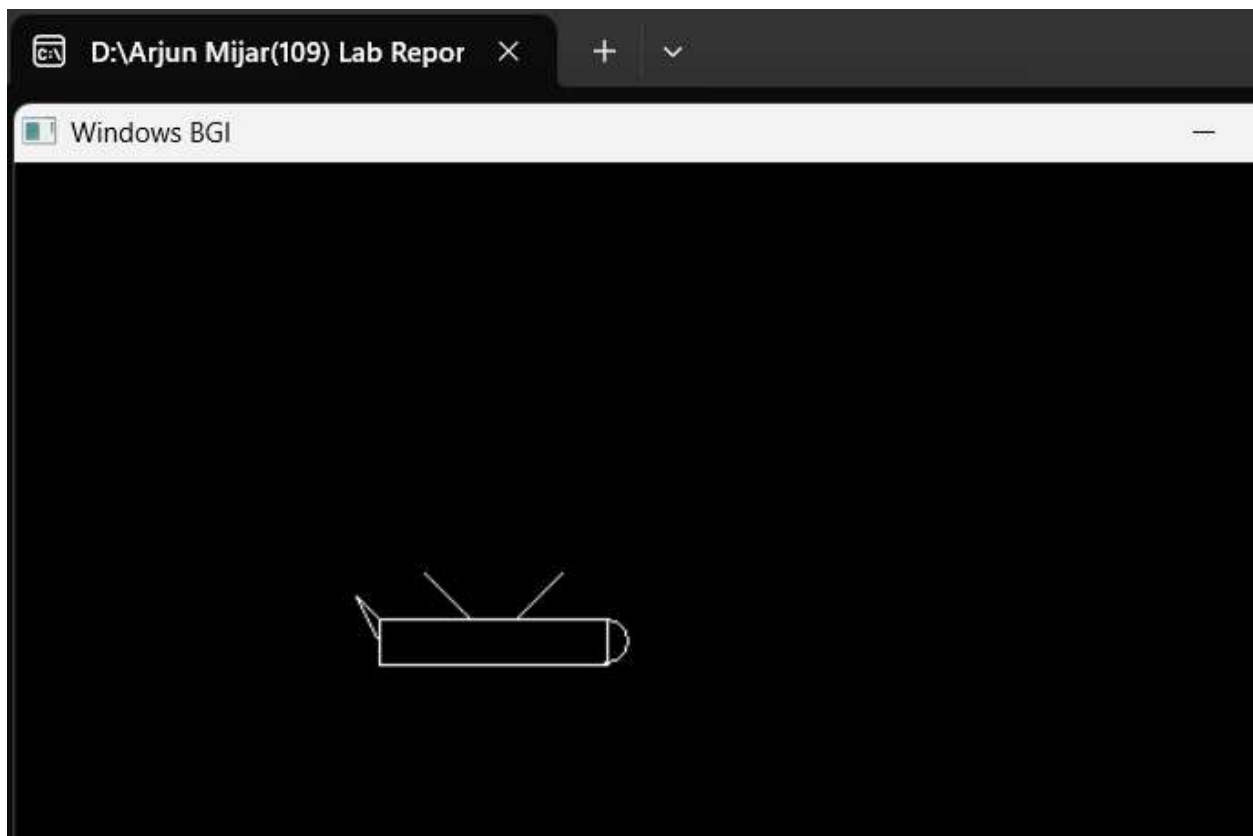
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "");

    for (int i = 0; i < getmaxx(); i += 5) {
        cleardevice();
        drawAeroplane(i, 200);
        delay(50);
    }

    getch();
    closegraph();
    return 0;
}

```

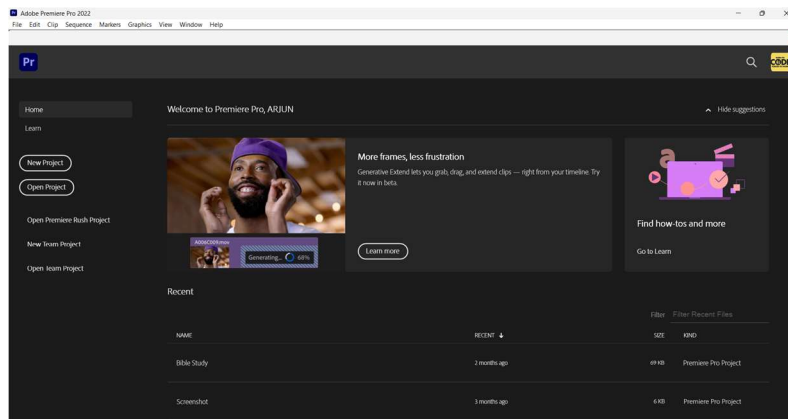
Output:



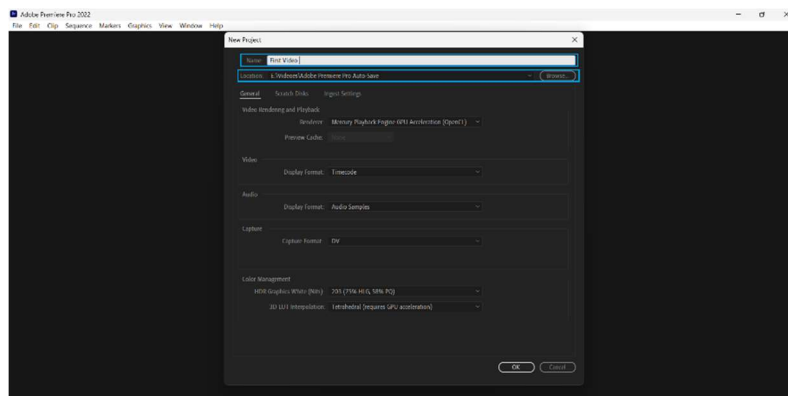
Lab 5: Write a program/ algorithm to import, edit and export a video on some video authoring software. E.g. Premier pro/ Rush

Import Video:

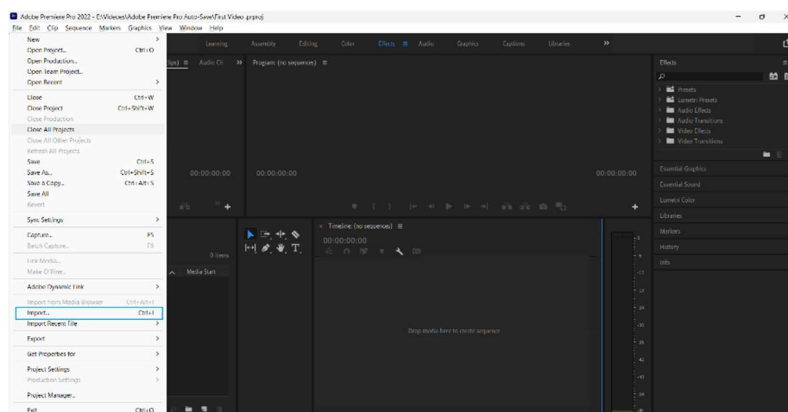
First, Open Adobe Premiere Pro.



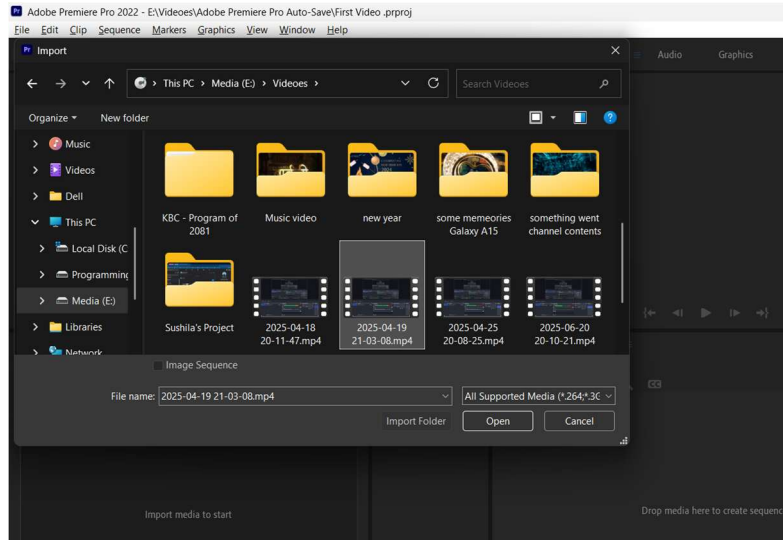
Create a New Project → Give it a name and set the location.



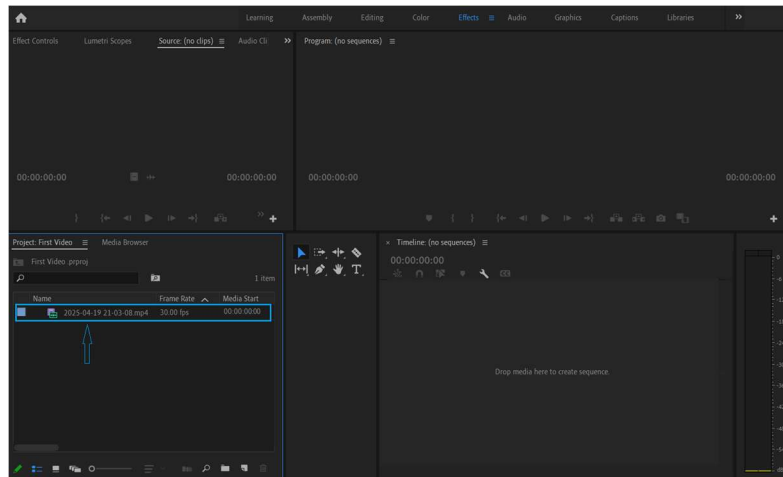
Go to File → Import, or press **Ctrl + I**.



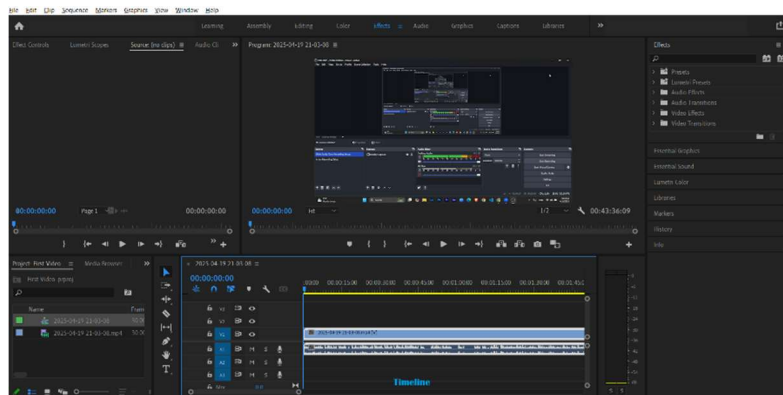
Select the video/audio/images from your PC.



The imported files appear in the Project Panel.

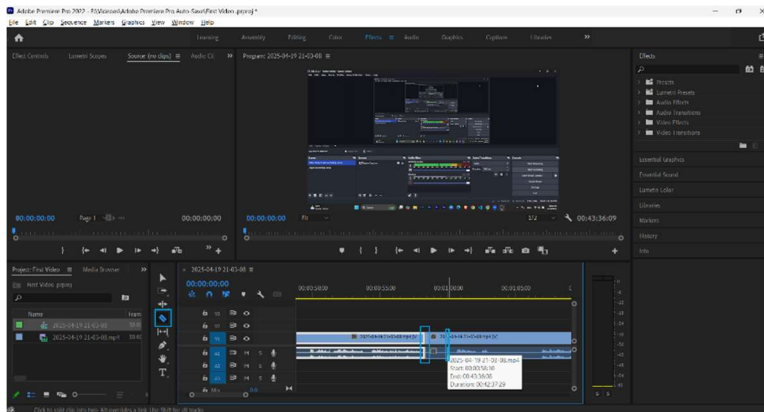


Drag your video to the Timeline to start editing.

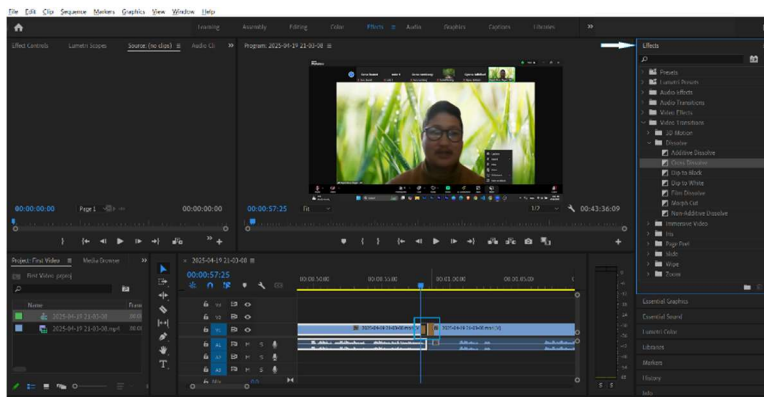


Edit Video:

Trim or cut clips: Use the Razor Tool (C) or drag clip edges in the timeline.

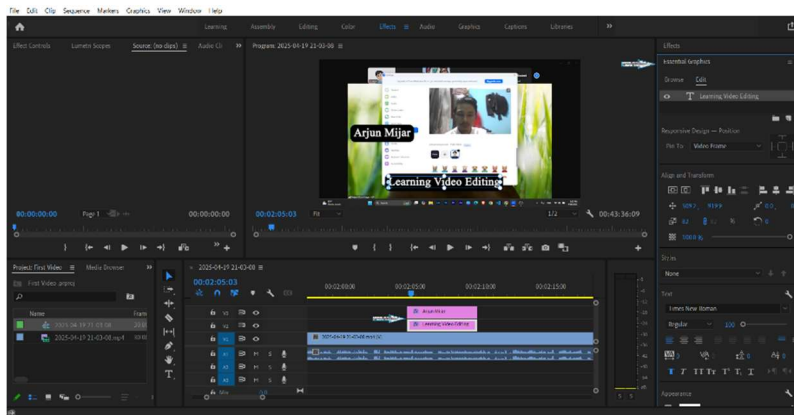


Add transitions: Go to Effects → Video Transitions and drag a transition between two clips.

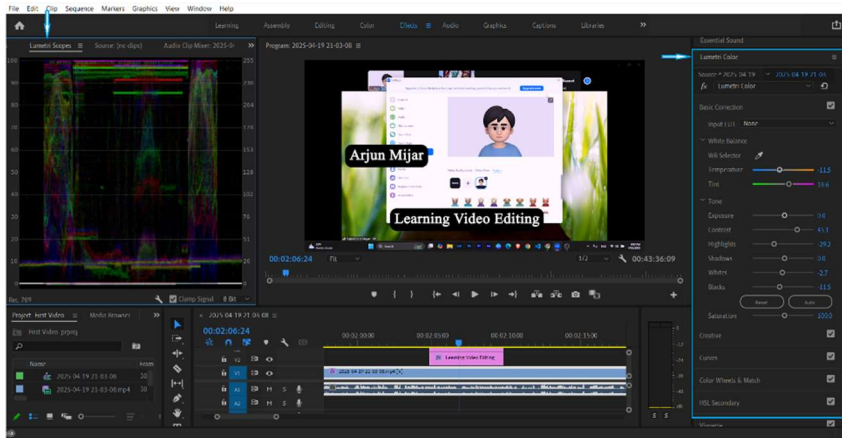


Add text/title:

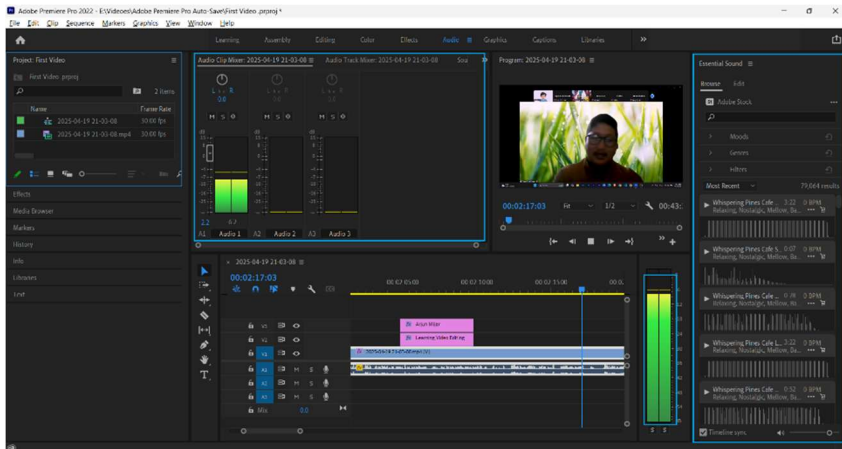
- Use Type Tool (T) to write on screen.
- Customize in the Essential Graphics Panel.



Apply effects: Drag from Effects Panel (like blur, color correction) to your clip.

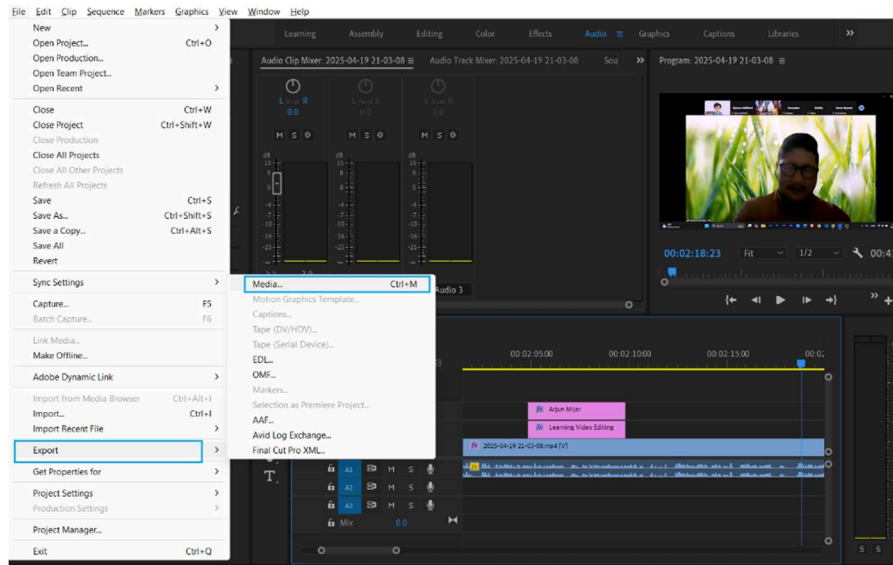


Audio editing: Adjust volume, add background music or voiceover.

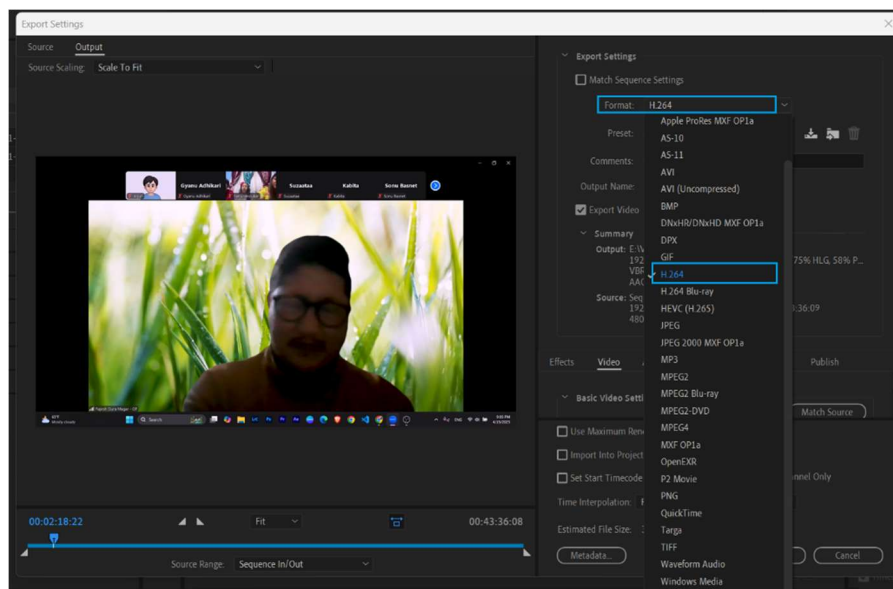


Export Video:

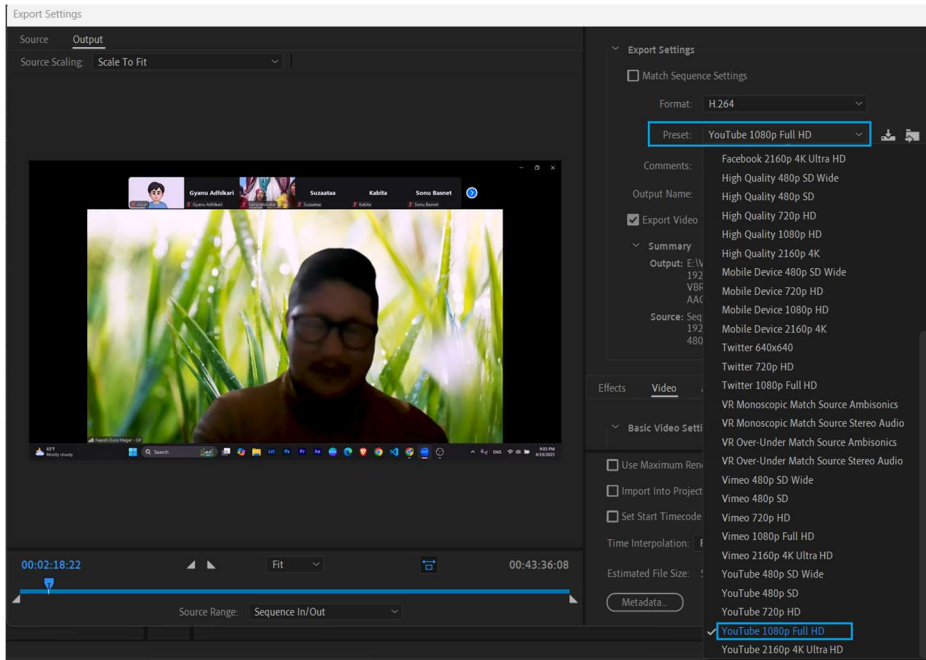
Go to File → Export → Media, or press `Ctrl + M`.



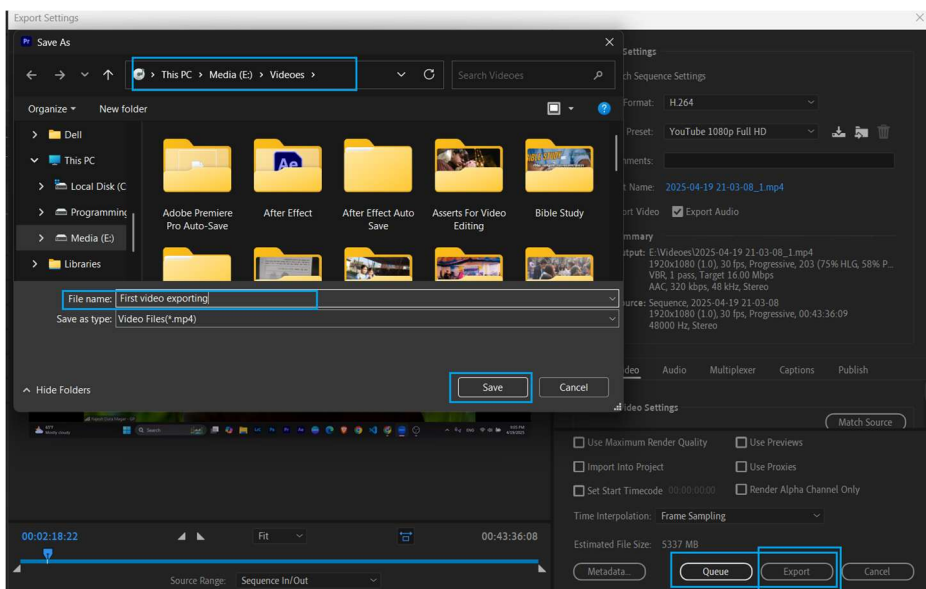
Set format to H.264 (for MP4 output).



Choose a Preset like "YouTube 1080p" or "Match Source - High Bitrate".



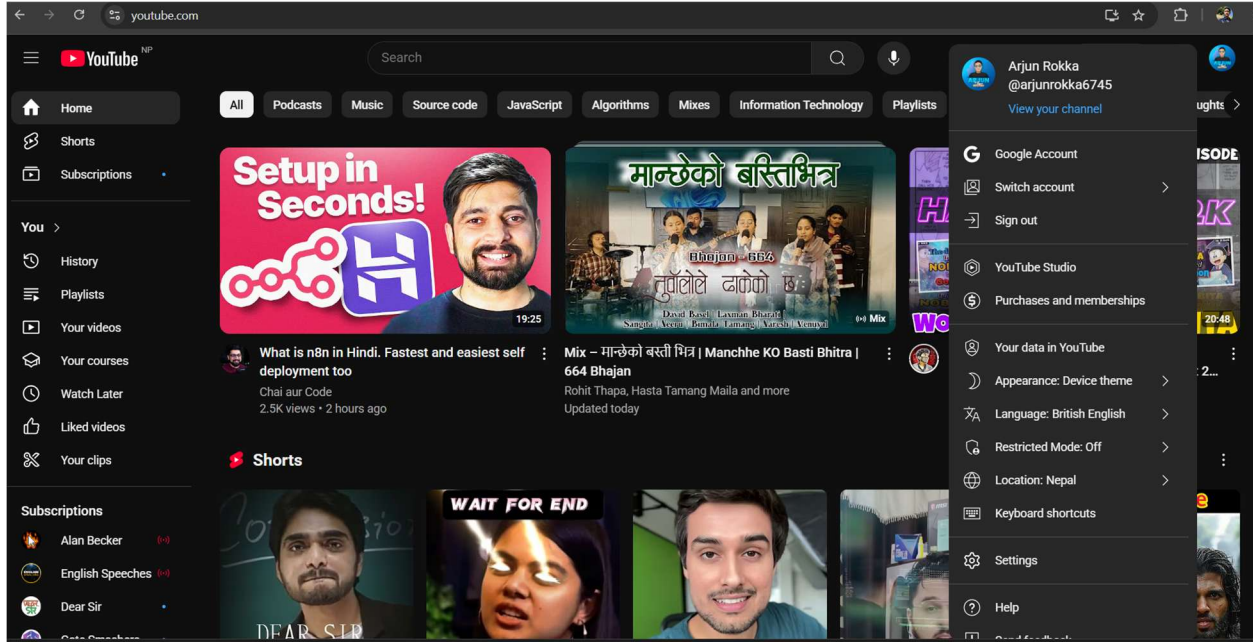
Set output name & location and Click Export or Queue (to render in Adobe Media Encoder).



Lab 6: Write a step by step program/ algorithm to upload the video to youtube and embed that video on a HTML page.

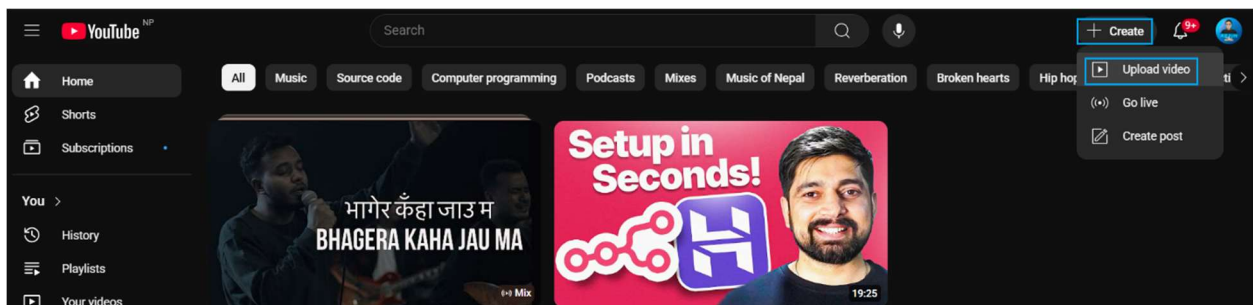
Step 1: Sign In

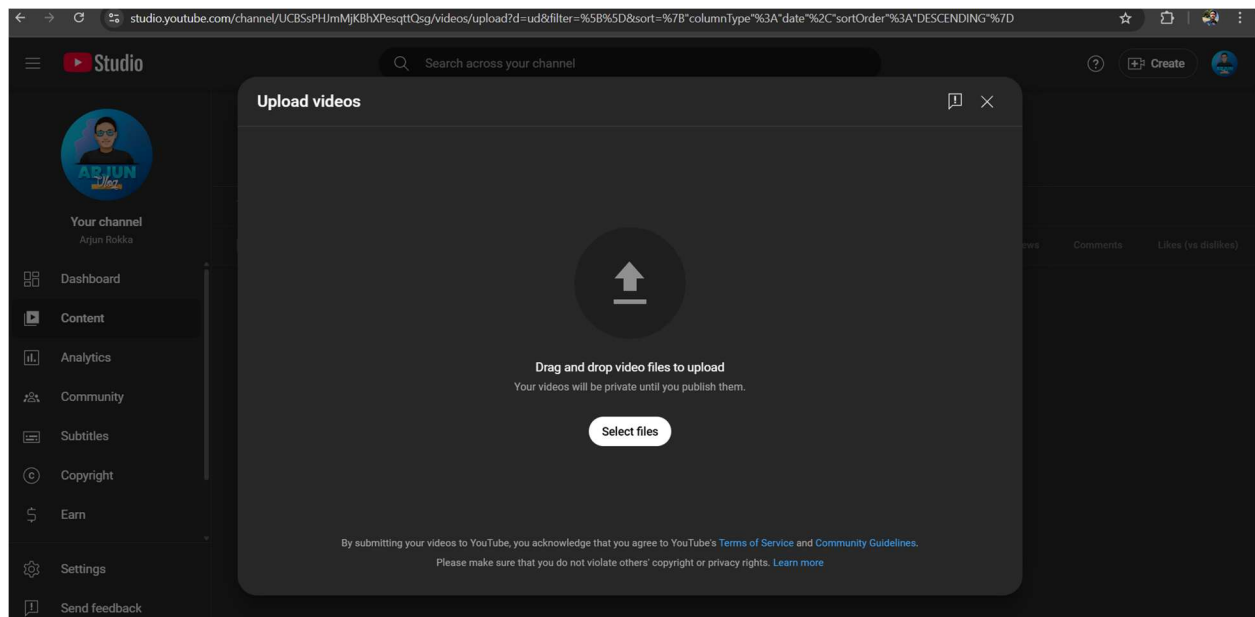
1. Go to <https://www.youtube.com>.
2. Click "Sign In" at the top right (if you're not already logged in).
3. Log in using your Google account.



Step 2: Open the Upload Page

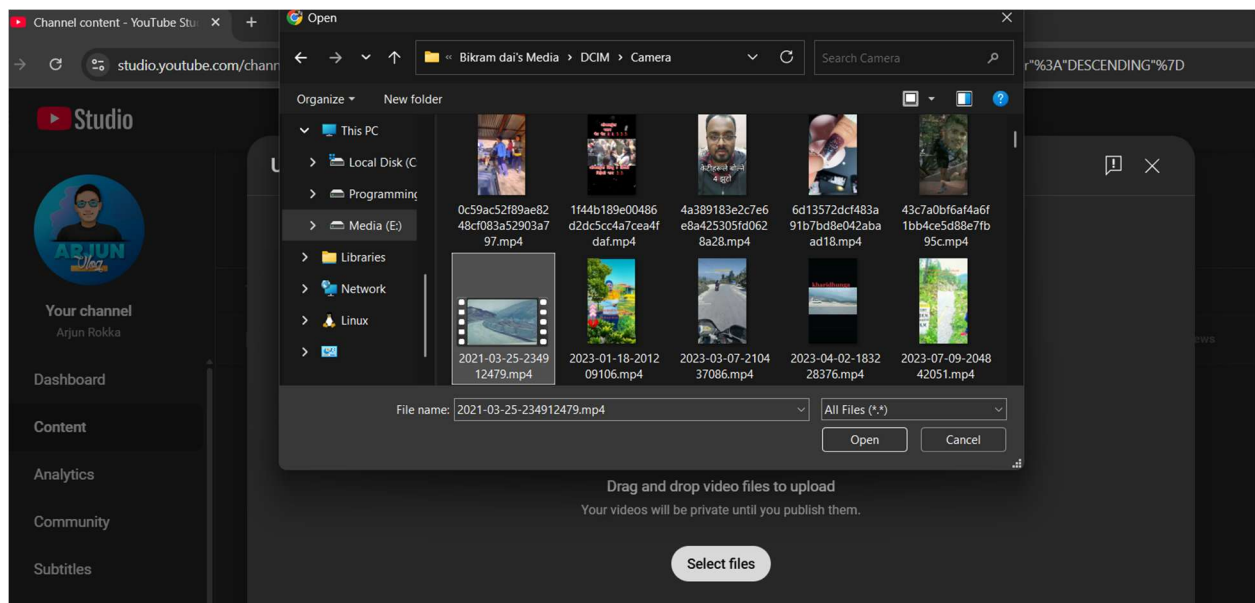
1. Click the camera icon with a "+" at the top right corner.
2. Select "Upload video".
 - o This will open YouTube Studio in a new tab.





Step 3: Select Video File

1. In YouTube Studio, click “Select Files”, or drag and drop your video.
2. Choose the video file (MP4, MOV, AVI, etc.) from your computer.



Step 4: Fill in Video Details

◇ Title:

- Give your video a meaningful title (max 100 characters).

◇ Description:

- Add info about the video, such as context, links, credits, etc.

◇ Thumbnail:

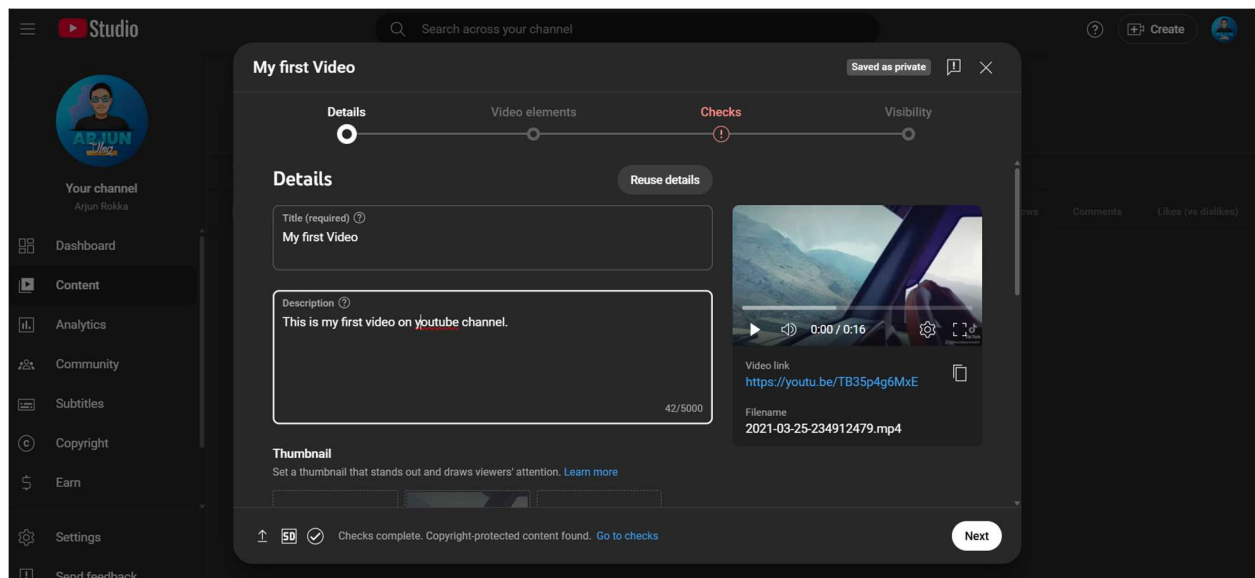
- You can upload a custom thumbnail or choose one auto-generated by YouTube.

◇ Playlist (optional):

- Add your video to an existing playlist or create a new one.

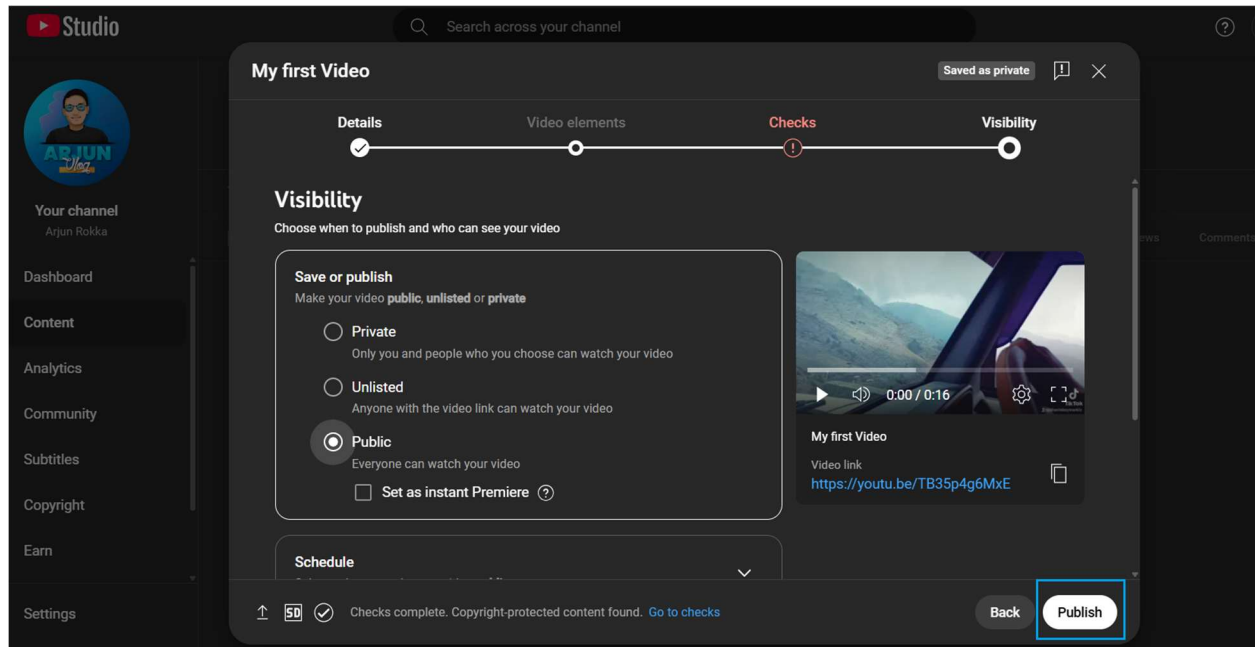
◇ Audience:

- Select **Yes** or **No** for “Is this video made for kids?”



Step 7: Publish

- After filling everything, click **“Save”** or **“Publish”**.
- Wait while the video uploads and processes.



Lab 7: Write a program to implement run length coding on given set of strings.

Source code:

```
#include <iostream>

using namespace std;

void printRLE (string str)
{
    int n = str.length() ;
    for (int i=0;i<n;i++){
        // Count occurrences of current character
        int count= 1;
        while (i < n-1 && str[i] == str[i+1]){
            count++;
            i++;
        }
        // Print character and its count
        cout<<str[i] <<count;
        }
    }

int main()
{
    cout<<"***By Arjun Mijar***"<<endl<<endl;
    string str;
    cout<<"Enter the string:";
    cin>>str;
    printRLE(str);
    return 0;
}
```

Output:

```
D:\Arjun Mijar(109) Lab Repor  ×  +  ∨  
***By Arjun Mijar***  
  
Enter the string:BBAACCABDEAAF  
B2A2C2A1B1D1E1A2F1  
-----  
Process exited after 49.82 seconds with return value 0  
Press any key to continue . . . |
```

Lab 8: Write a program to implement Huffman coding on some given characters and their frequencies.

Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

// Node structure
typedef struct node {
    char ch;
    int freq;
    struct node *left;
    struct node *right;
} node;

// Heap array and size
node *heap[100];
int heapSize = 0;

// Function to insert a node into the min-heap
void Insert(node *element) {
    heapSize++;
    heap[heapSize] = element;
    int now = heapSize;
    while (heap[now / 2]->freq > element->freq) {
        heap[now] = heap[now / 2];
        now /= 2;
    }
    heap[now] = element;
```

```
}
```

```
// Function to delete and return the node with minimum frequency
```

```
node *DeleteMin() {  
    node *minElement = heap[1];  
    node *lastElement = heap[heapSize--];  
    int now, child;  
    for (now = 1; now * 2 <= heapSize; now = child) {  
        child = now * 2;  
        if (child != heapSize && heap[child + 1]->freq < heap[child]->freq) {  
            child++;  
        }  
        if (lastElement->freq > heap[child]->freq) {  
            heap[now] = heap[child];  
        } else {  
            break;  
        }  
    }  
    heap[now] = lastElement;  
    return minElement;  
}
```

```
// Recursive function to print Huffman codes
```

```
void print(node *temp, char *code) {  
    if (temp->left == NULL && temp->right == NULL) {  
        printf("char %c : code %s\n", temp->ch, code);  
        return;  
    }  
}
```

```

    int length = strlen(code);
    char leftcode[20], rightcode[20];
    strcpy(leftcode, code);
    strcpy(rightcode, code);
    leftcode[length] = '0';
    leftcode[length + 1] = '\0';
    rightcode[length] = '1';
    rightcode[length + 1] = '\0';

    print(temp->left, leftcode);
    print(temp->right, rightcode);
}

// Main function
int main() {
    heap[0] = (node *)malloc(sizeof(node));
    heap[0]->freq = 0;

    int n;
    printf("Enter the number of characters: ");
    scanf("%d", &n);

    printf("Enter characters and their frequencies:\n");

    char ch;
    int freq;

    for (int i = 0; i < n; i++) {

```

```

scanf(" %c %d", &ch, &freq);
node *temp = (node *)malloc(sizeof(node));
temp->ch = ch;
temp->freq = freq;
temp->left = temp->right = NULL;
Insert(temp);
}

```

```

// Special case: only one character
if (n == 1) {
    printf("char %c : code 0\n", heap[1]->ch);
    return 0;
}

```

```

// Build Huffman tree
for (int i = 0; i < n - 1; i++) {
    node *left = DeleteMin();
    node *right = DeleteMin();
    node *temp = (node *)malloc(sizeof(node));
    temp->ch = 0;
    temp->freq = left->freq + right->freq;
    temp->left = left;
    temp->right = right;
    Insert(temp);
}

```

```

node *tree = DeleteMin();
char code[20];

```

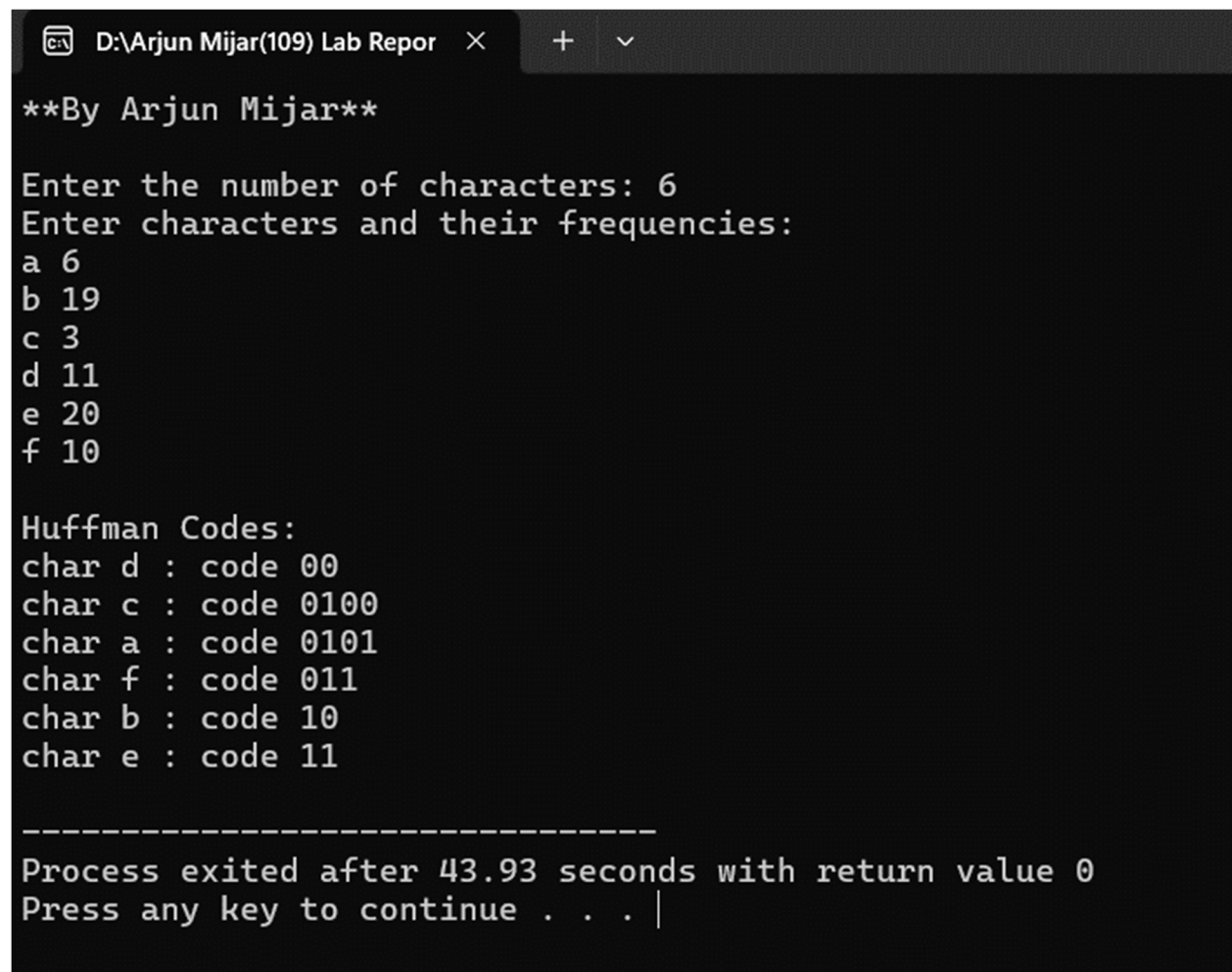


```
code[0] = '\0';

printf("\nHuffman Codes:\n");
print(tree, code);

return 0;
}
```

Output:



```
D:\Arjun Mijar(109) Lab Repor × + v

**By Arjun Mijar**

Enter the number of characters: 6
Enter characters and their frequencies:
a 6
b 19
c 3
d 11
e 20
f 10

Huffman Codes:
char d : code 00
char c : code 0100
char a : code 0101
char f : code 011
char b : code 10
char e : code 11

-----
Process exited after 43.93 seconds with return value 0
Press any key to continue . . . |
```